

1. Administrative & Study Identification

Full Study Title: A Study of Atezolizumab (Tecentriq) in Combination With Bevacizumab to Investigate Safety and Efficacy in Patients With Unresectable Hepatocellular Carcinoma Not Previously Treated With Systemic Therapy-Amethista **Short Title/ Acronym:** AMETHISTA **Protocol Number:** ML42243 **NCT Number:** NCT04487067 **Trial Status:** COMPLETED **Sponsor Name:** Roche **Investigational Product:** Atezolizumab + Bevacizumab **Phase of Development:** Phase 3 (Phase IIIB) **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of the atezolizumab plus bevacizumab combination, specifically by assessing the incidence of grade 3-5 bleeding/hemorrhages in patients with unresectable Hepatocellular Carcinoma (HCC) who have received no prior systemic treatment. **Secondary Objectives:** To evaluate the effectiveness of the treatment, including Overall Survival (OS) and Progression-Free Survival (PFS), as well as track the Number of Participants With Treatment-Emergent Adverse Events (TEAEs).

2.2 Background & Rationale To address the need for real-world evidence and evaluate the safety and effectiveness of atezolizumab and bevacizumab in a Western (Italian) population of patients with systemic treatment-naïve advanced and unresectable HCC. Because the etiology of background liver disease can differ significantly between Eastern and Western populations, this study validates the use of this combination therapy as a standard of care for advanced HCC in this demographic, complementing the findings of the pivotal IMbrave150 trial.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 152 participants **Number of Sites:** 21 Italian oncology and gastroenterology centers **Estimated Study Start:** 25-Aug-2020 **Estimated Primary Completion:** 13-Aug-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Unresectable HCC with diagnosis confirmed by histology, with a biopsy within 6 months from recruitment. 2. Disease that is not amenable to curative surgical and/or locoregional therapies, or progressive disease after surgical and/or locoregional therapies. 3. No prior systemic therapy for HCC. 4. At least one measurable untreated lesion. 5. Patients who received prior local therapy are eligible provided the target lesion(s) have not been previously treated with local therapy or the target lesion(s) within the field of local therapy have subsequently progressed in accordance with RECIST version 1.1. 6. ECOG Performance Status of 0 or 1 within 7 days prior to recruitment. 7. Child-Pugh class A within 7 days prior to recruitment. 8. Patients must undergo an esophagogastroduodenoscopy (EGD), and all size of varices (small to large) must be assessed. In case of varices at high risk of bleeding (corresponding to medium [F2] or large [F3] varices, or F1 varices with cherry red spots or red wale marking), prophylactic treatment per local standard of care must be adopted prior to enrollment. Patients who underwent an EGD within 6 months prior to study initiation do not need to repeat if they had no high-risk varices. 9. Adequate hematologic and end-organ function. 10. Resolution of any acute, clinically significant treatment-related toxicity from prior therapy to Grade ≤ 1 prior to study entry, with the exception of alopecia. 11. Negative HIV test at screening (exception: patients with a positive HIV test are eligible if they are stable on anti-retroviral therapy, have a CD4 count $\geq 200/\mu\text{L}$, and an undetectable viral load). 12. In patients with viral HCC, documented virology status of hepatitis, confirmed by screening HBV and HCV serology test. 13. For women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive methods, and agreement to refrain from donating eggs. 14. For men: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive measures, and agreement to refrain from donating sperm.

4.2 Exclusion Criteria 1. History of leptomeningeal disease or brain metastases; symptomatic, untreated, or actively progressing central nervous system (CNS) metastases. 2. Active or history of autoimmune disease or immune deficiency. 3. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active pneumonitis on screening chest computed tomography (CT) scan. 4. Known active tuberculosis. 5. Significant cardiovascular disease within 3 months prior to initiation of study treatment, unstable arrhythmia, or unstable angina. 6. History of malignancy other than HCC within 5 years prior to screening (except malignancies with a negligible risk of metastasis or death). 7. Prior allogeneic stem cell or solid organ transplantation. 8. Pregnancy or breastfeeding, or intention of becoming pregnant during study

treatment or within at least 5 months after the last dose of atezolizumab and 6 months after the last dose of bevacizumab. 9. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 10. Untreated or incompletely treated esophageal and/or gastric varices with bleeding or high-risk for bleeding; prior bleeding event due to oesophageal and/or gastric varices within 6 months prior to study treatment. 11. Clinically evident ascites, uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures. 12. Co-infection of HBV and HCV; co-infection with HBV and hepatitis D viral infection. 13. Clinically significant uncontrolled or symptomatic hypercalcemia. 14. Inadequately controlled arterial hypertension. 15. Significant vascular disease within 6 months prior to initiation of study treatment. 16. History of haemoptysis; evidence of bleeding diathesis or significant coagulopathy. 17. History of gastrointestinal (GI) fistula, GI perforation, or intra-abdominal abscess within 6 months prior to study treatment. 18. History of intestinal obstruction and/or clinical signs or symptoms of GI obstruction including sub-occlusive disease related to the underlying disease or requirement for routine parenteral hydration, parenteral nutrition, or tube feeding prior to study treatment. 19. Metastatic disease that involves major airways or blood vessels, or centrally located mediastinal tumor masses of large volume. 20. Local therapy to liver within 28 days prior to initiation of study treatment or non-recovery from side effects of any such procedure.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product 1:

Atezolizumab (1200 mg every 3 weeks, IV) **Investigational Product**

2: Bevacizumab (15 mg/kg every 3 weeks, IV) *Bevacizumab Details:*

- **ATC Codes:** S01LA08, L01FG01 - **Trade Names:** Avastin, Vegzelma, Alymsys, Zirabev, Mvasi **Duration of Treatment:** Until unacceptable toxicity, loss of clinical benefit, or patient refusal.

5.2 Concomitant & Prohibited Therapy Permitted: Prophylactic treatment per local standard of care for esophageal/gastric varices prior to enrollment; anti-retroviral therapy for HIV-positive patients (if stable and CD4 $\geq 200/\mu\text{L}$).

Prohibited: Any prior systemic therapies for HCC.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, EGD (if not done within 6 months), ECOG, Child-Pugh assessment
Baseline	Day 0	First Dose of Atezolizumab + Bevacizumab, Baseline Tumor Assessment
Interim	Every 3 Weeks	Safety and Efficacy Assessments, IV infusion of study

drugs | | **End of Study** | Up to 48 months | Final survival and progression follow-up (OS and PFS tracking) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Number of Participants With Grade 3-5 National Cancer Institute Common Terminology Criteria for Adverse Events, Version 5.0 (NCI CTCAE V5) Bleeding/Haemorrhage.

Definition: An adverse event (AE) was any untoward medical occurrence in a clinical investigation participant administered a pharmaceutical product, regardless of causal attribution. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product. Severity of AEs was graded using NCI CTCAE v5.0. - Grade 3 = Severe/medically significant but not immediately life-threatening, hospitalization/prolongation of hospitalization indicated; disabling; limiting self-care activities of daily living. - Grade 4 = Life-threatening consequences, urgent intervention indicated. - Grade 5 = Death related to AE.

7.2 Secondary Endpoints 1. Overall Survival (OS) 2. Number of Participants With Treatment-Emergent Adverse Events (TEAEs) 3. Progression-free Survival (PFS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of Treatment-Emergent Adverse Events (TEAEs) with a specific focus on bleeding/hemorrhage events associated with bevacizumab.

Serious Adverse Events (SAEs): Monitored continuously and reported per standard safety guidelines, including targeted monitoring for Grade 3-5 hemorrhages. **Data Monitoring Committee (DMC):** Managed by the sponsor (Roche) to evaluate the safety profile throughout the study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population (N=152) for effectiveness outcomes; Safety Set (N=149 treated) for all AE/SAE evaluations. **Sample Size Justification:** A sample of 152 patients was appropriately sized to evaluate the safety incidence (bleeding) and effectiveness in the Italian advanced HCC population. **Handling of Missing Data:** The Kaplan-Meier (K-M) method was used to estimate time-to-event outcomes (OS, PFS); patients who were alive or had not progressed were censored at the last assessment date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard onsite and remote monitoring at the 21 participating centers. **Audit Trail:** Robust eCRF tracking for tumor responses (per RECIST v1.1) and rigorous validation of EGD records prior to enrollment.

11. Study Identifiers & Governance

Posting ID: 679311232348664716 **Primary ID:** ML42243 **Registry Number:** NCT04487067 **Sponsor/Lead Organization:** Roche **Study Title:** AMETHISTA **Official Regulatory Title:** A Phase IIIB, Single Arm, Multicenter Study of Atezolizumab (Tecentriq) in Combination With Bevacizumab to Investigate Safety and Efficacy in Patients With Unresectable Hepatocellular Carcinoma Not Previously Treated With Systemic Therapy-Amethista

12. Clinical Framework (Hepatocellular Carcinoma Specific)

Disease Area: Carcinoma, Hepatocellular **Preferred UMLS Name:** Hepatocellular Carcinoma **Semantic Type:** Neoplastic Process

Condition Definition: A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas.

MeSH Code: D002277 **HPO Code:** HP:0030731 **SNOMED ID:** 1187425009

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced/unresectable HCC **Education Protocol (HCP):** Site initiation and adverse event management training, specifically recognizing and managing variceal bleeding risks **Education Protocol (Patient):** Onsite training for treatment compliance, recognition of hemorrhage symptoms, and AE reporting **Quality Audit Mechanism:** Routine safety monitoring and objective tumor assessments utilizing RECIST v1.1

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 48 months **Onsite Visit Cadence:** Every 3 weeks (Q3W) concurrent with IV infusions **Remote Monitoring Frequency:** Follow-up for overall survival post-

progression as per protocol **Checklist Review Interval:** Every 3 weeks prior to each treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	07-Apr-2026				
Clinical Operations Lead	[Name]	_____	07-Apr-2026		Lead		
Statistician	[Name]	_____	07-Apr-2026				